

Lecture 1

Language of Data



Announcements

- ❖ Lab 0 on Wednesday
 - Getting our computers set up
 - Bring your computer to lab!
- ❖ Test out attendance link
 - Regular attendance starts next week

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 - Biometric readings, images, video
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- ❖ Examples of data:
 - Survey responses, quarterly sales figures, text messages
 - Biometric readings, images, video
 - This slide deck
- ❖ Basically anything that can be stored can be thought of as data

Structure in Data

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- ❖ We will focus on data that can be represented by rows and columns
 - A datum is often a row in this model
 - Called *tabular data*

Structure in Data

- ❖ *Unstructured data* is roughly everything else
- ❖ Common types:
 - Text (emails, pdfs, text messages)
 - Images, Audio, and Video
 - Chat logs, meeting minutes
 - Sensor readings from smart devices

Tabular Data

- ❖ Data format: rows and columns
- ❖ Each column is a *variable* (or *feature* or *attribute*)
- ❖ Each row is an *observation* (or *record*)
- ❖ The *unit of observation* is what an observation represents

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- ❖ Each column is a *variable* (or *feature* or *attribute*)
- ❖ Each row is an *observation* (or *record*)
- ❖ The *unit of observation* is what an observation represents
 - Data: class survey → Unit of observation: individual student
 - Data: weekly class survey → Unit of observation: individual student at given week

Variables in Tabular Data

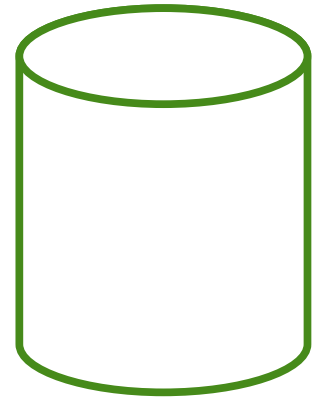
- ❖ Quantitative: *numerical* variables
 - Examples: age, exam score, height, years of education
- ❖ Qualitative: *categorical* variables
 - Examples: occupation, income (>\$50k), book cover
 - If categories can be ordered, we say the variable is *ordinal*

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 - If categories can be ordered, we say the variable is *ordinal*
 - Categories may encode complex relations such as “state” or “county”
- ❖ Easy check: does the mean make sense?

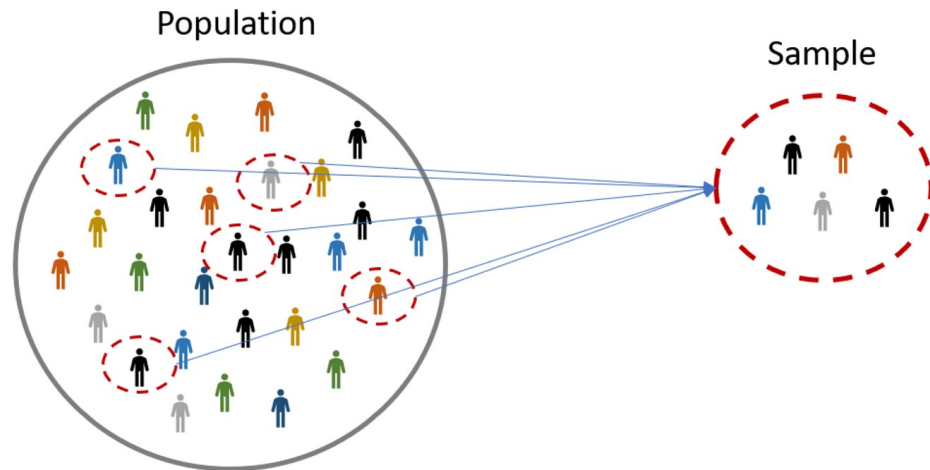
Data Provenance

- ❖ Where did this data come from?
- ❖ How was it collected?
- ❖ When was it collected?
- ❖ How are the variables coded?
- ❖ Why does this matter for analyzing the data?



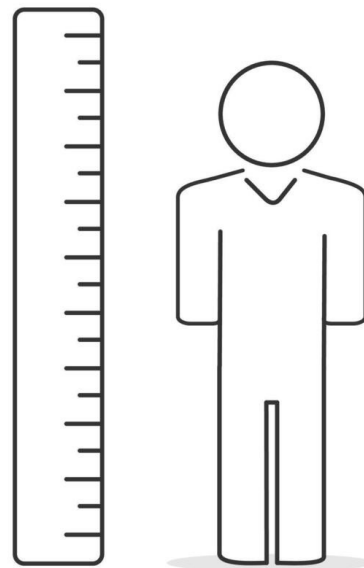
Sample vs. Population

- ❖ We can think of our data as a *sample* from some *population*
- ❖ We often want to learn about the population from the sample
 - This is called *inference*
 - An inference that generalizes from the sample to the population is *valid*
- ❖ Studying the provenance of data can tell us if an inference is valid



Closing Thoughts

- ❖ Imagine you're given a dataset containing the heights of some Allegheny students
- ❖ You calculate the mean height
- ❖ Is this mean a good estimate of the average height of Allegheny students?
- ❖ When might it be? When might it not?



Next Time

- ❖ Read CIT Ch. 2
- ❖ Forming hypotheses
- ❖ Correlation vs. Causation